BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type

Q1. Which among following oxides is amphoteric:

(A) Na2O
(B) MgO
(C) SO3
(D) ZnO

Q2. Chile saltpetre has the chemical formula:

(A) NaNO3
(B) KNO3
(C) Na2B4O7
(D) Na2CO3

Q3. Tincal is a mineral of:

(A) Al
(B) B
(C) Si
(D) C

Q4. Laughing gas is chemically:

(A) NO
(B) N2O
(C) NO2
(D) N2O4

Q5. The anhydride of $HClO_4$ is:

(A) ClO_3 (B) ClO_2

(C) Cl_2O_5 (D) Cl_2O_7

Q6. Which of the following is a typical transition element:

(A) Sc (B) Y

(C) Ra (D) Co

Q7. The state of hybridization of carbon atom in methane is:

(A) sp^3 (B) sp^2

(C) sp (D) dsp^2

Q8. Formula of chloroform is:

(A) CH_3Cl (B) CCl_4

(C) CH_2Cl_2 (D) $CHCl_3$

Q9. The electrophile in aromatic sulphonation is:

(A) H_2SO_4 (B) HSO_4^-

(C) SO_3 (D) SO_3^+

Q10. Which one of the following is not a nucleophile:

(A) H_2O (B) H_2S

(C) BF_3 (D) NH_3

Q11. Which compound is called a universal solvent:

(A) H_2O (B) CH_3OH

(C) C_2H_5OH (D) $CH_3\text{-}O\text{-}CH_3$

Q12. The carbon atom of a carboxyl group is:

(A) sp hybridized (B) sp^2 hybridized

(C) sp^3 hybridized (D) sp^3d^2 hybridized

Q13. Acetic acid is manufactured by:

(A) Distillation (B) Fermentation

(C) Ozonolysis (D) Esterification

Q14. The reaction between fat and NaOH is called:

(A) Esterification (B) Hydrogenolysis

(C) Fermentation (D) Saponification

Q15. Which is not a calcareous material:

(A) Lime
(B) Clay
(C) Marble
(D) Marine shell

Q16. Residence time of NO in atmosphere is:

(A) 1 day
(B) 2 days
(C) 3 days
(D) 4 days

Q17. Ozone layer is present in range of:

(A) 0 - 5 km
(B) 10 - 15 km
(C) 15 - 25 km
(D) 25 - 28 km

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

> Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Lanthanides contraction controls the atomic sizes of element of 6th and 7th period. Comment.

(ii) Why is the oxidation state vary in a period but remain almost constant in a group?

(iii) Point out the two advantages of Down's cell.

(iv) Why 2% Gypsum is added in the cement?

(v) Give two uses of silicones.

(vi) Give any four uses of silicates.

(vii) Why does aqua regia dissolve gold and platinum?

(viii) Describe ring test for the conformation [sic, print for 'confirmation'] of the presence of nitrate ions in solution.

(ix) What is chromyl chloride test?

(x) Write down the any [sic] two uses of potassium dichromate.

(xi) Describe the composition of Portland cement.

(xii) What are phosphatic fertilizers? Give one example.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) What is disproportionation reaction? Give example.

(ii) What are Freon and Teflon?

(iii) What is carbonization of coal?

(iv) What is steam cracking?

(v) 1-alkyne behave as weak acid. Give reason.

(vi) How will you convert methane into ethane?

(vii) Give mechanism of Kolb'e [sic] electrolytic method for the preparation of ethane.

(viii) Why dry ether is used for the preparation of Grignard's reagent?

(ix) Differentiate between E1 and E2 reaction with any two points.

(x) Define co-polymer. Give example.

(xi) What is meant by iodine number?

(xii) Differentiate between DNA and RNA.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Write down the structural formula of two fused rings aromatic hydrocarbons (Two examples).

(ii) What is Wurtz-Fitting reaction? Give an example.

(iii) Describe one method for preparing phenol.

(iv) Why ethanol has higher boiling point than diethyl ether?

(v) How does acetaldehyde reacts with: (i) C2H5MgI (ii) K2Cr2O7/H2SO4

(vi) How acetic acid is prepared from methyl nitrile?

(vii) How amino acids are prepared by the stretcher [sic, print for 'Strecker'] synthesis?

(viii) How is oil spillage affecting marine life?

(ix) How water is purified by aeration?

> Note: SECTION-II

> Note: Note: Attempt any THREE questions.

Q5. (Marks 4) (a) What are the improvements in the Mendeleev table?

(b) What are silicates? How sodium silicate is prepared? Give its two uses. 4

Q6. (Marks 4) (a) Explain the peculiar behaviour of Lithium, give any eight points.

(b) Discuss the following properties of transition elements: (i) Paramagnetism (ii) Colour. 4

Q7. (Marks 4) (a) Describe the structure of ethene by sp2 hybridization.

(b) Discuss briefly the two possible mechanisms of beta-elimination reaction. Write any four differences of E2 and E1 reaction. 4

Q8. (Marks 4) (a) Convert acetylene to : (i) Vinyl acetylene (ii) Divinyl acetylene (iii) Benzene (iv) Acetaldehyde.

(b) Describe these test : (i) Fehling's solution test. (ii) Tollen's test. 4

Q9. (Marks 4) (a) Describe the structure of Benzene on the basis of atomic orbital treatment.

(b) Explain the acidic behaviour of phenol. 4

GROUP – II

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type

Q1. If the salt bridge is not used between two half cells, then the voltage:

(A) Decrease rapidly
(B) Decrease slowly
(C) Does not change
(D) Drops to zero

Q2. With increase of 10C temperature. The rate of reaction doubles. This is [stem partially cut in source scan]:

(A) Decrease in activation energy of reaction
(B) Decrease in the number of collisions between reactant molecules
(C) Increase in activation energy of reactants
(D) Increase in number of effective collisions

Q3. The number of moles of CO2 which contain 8.0 g of oxygen:

(A) 0.25
(B) 0.50
(C) 1.0
(D) 1.50

Q4. During process of crystallization, the hot saturated solution:

(A) Is cooled very slowly to get large sized crystals

(B) Is cooled at a moderate rate to get medium sized crystals

(C) Is evaporated to get the crystals of the product

(D) Is mixed with an immiscible liquid to get the pure crystals of the product

Q5. Compound which undergo sublimation is:

(A) KMnO4

(B) CaCO3

(C) NH4Cl

(D) Na2CO3

Q6. If absolute temperature of a gas is doubled and the pressure is reduced to one half, the volume of the gas will be:

(A) Remains unchanged

(B) Increase four times

(C) Reduce to 1/4

(D) Be doubled

Q7. The pressure remaining constant; at which temperature the volume of a gas will become twice of its volume:

(A) 546C

(B) 200C

(C) 546 K

(D) 273 K

Q8. Acetone and chloroform are soluble in each other due to:

(A) Intermolecular hydrogen bonding

(B) Ion-dipole interaction

(C) Instantaneous dipole

(D) All of the above

Q9. Which of the following is a pseudo solid?

(A) CaF2

(B) Glass

(C) NaCl

(D) All

Q10. When 6d orbital is complete, the entering electron goes into:

(A) 7f

(B) 7s

(C) 7p

(D) 7d

Q11. Rutherford's model of atom failed because:

(A) The atom did not have a nucleus and electron

(B) It did not account for the attraction between proton and neutron

(C) It did not account for the stability of the atom

(D) There is actually no space between the nucleus and the electrons

Q12. Which of the following statement is not correct regarding bonding molecular orbitals?

(A) Bonding molecular orbitals possess less energy than atomic orbitals from which they are formed

(B) Bonding molecular orbitals have low electronegativity between the two nuclei

(C) Every electron in the bonding molecular orbitals contribute to the attraction between atoms

(D) Bonding molecular orbitals are formed when the electron wave undergo constructive interference

Q13. In methanol, bond between carbo [sic, print for 'carbon'] and oxygen:

(A) Ionic

(B) Non-polar

(C) Polar

(D) Coordinate

Q14. The change in heat contents of a chemical reaction at constant temperature and pressure is called:

(A) Enthalpy change

(B) Heat of sublimation

(C) Bond energy

(D) Internal energy change

Q15. Which statement about the following equilibrium is correct: 2SO2(g) + O2(g) reversible 2SO3(g) delta H = -188.3 KJ mole-1:

(A) The value of Kp falls with a rise in temperature

(B) The value of Kp falls with increasing pressure

(C) Adding V2O5 catalyst increase the equilibrium yield of sulphur trioxide

(D) The value of Kp is equal to Kc

Q16. 18 g glucose is dissolved in 90 g of water. The relative lowering of vapour pressure is equal to:

(A) 1/5

(B) 5.1

(C) 1/51

(D) 6

Note: Question numbering on the printed objective paper skips 3: the paper prints Q.1, Q.2, then Q.4-Q.17. Transcribed here renumbered 1-16 as printed in sequence.

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Lanthanides contraction controls the atomic sizes of elements of 6th and 7th periods. Give reason.

(ii) Explain the variations in melting points along the short periods.

(iii) Why is the aqueous solution of Na2CO3 alkaline in nature? Give reactions only.

(iv) What happens when LiOH is heated to red hot? Give reaction.

(v) How does boric acid react with: (a) Ethyl alcohol (B) [sic, printed capital B] Na2CO3

(vi) Give chemistry of borax-bead test.

(vii) Name the allotropes of phosphorus. How red phosphorus is prepared?

(viii) Give the reaction of phosphorus with : (a) Thionyl chloride (b) Cl2(g)

(ix) Briefly discuss the property of paramagnetism in transition elements compounds.

(x) What are : (a) Interstitial compounds (b) Substitutional alloys.

(xi) Briefly discuss the digestion process in paper manufacturing.

(xii) Discuss the reactions taking place in first 24 hours of setting of cement.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) HF is weaker acid than HCl?

(ii) What are freons and teflon?

(iii) Discuss carbonization of coal.

(iv) Define functional group. Give one example.

(v) Why is sigma bond inert?

(vi) Discuss hydroxylation of ethene.

(vii) How is water added to propyne? Write reaction.

(viii) How are tetramethyl and tetraethyl lead prepared?

(ix) How can Grignard reagent be prepared? [middle of question watermarked, uncertain]

(x) Define polysaccharides, also give example.

(xi) Differentiate between DNA and RNA.

(xii) What is hardening of olls [sic, print for 'oils']? Give reaction.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) What is Wurtz-Fitting reaction.

(ii) What are polycyclic aromatic hydrocarbons? Give two examples.

(iii) Convert phenol into: (a) 2,4,6 tribromophenol (ii) [sic, printed 'ii'] Cyclohexanol.

(iv) Write equations for the preparation of ethanol by the fermentation of Molasses.

(v) What is iodoform test?

(vi) Convert acetylene into acetic acid.

(vii) Write down the structural formulas of Glycine and Alanine amino acids.

(viii) How determents [sic, print for 'detergents'] are threat to aquatic animal life?

(ix) Write down the four harmful effects of acid rain.

> Note: SECTION-II

> Note: Note: Attempt any THREE questions.

Q5. (Marks 4) (a) Write down two points of similarities and two points of differences between hydrogen and halogen (VII A).

(b) Give one method for the preparation of H3BO3. How does it react with : (i) NaOH (ii) Na2CO3 (iii) C2H5OH 4

Q6. (Marks 4) (a) Discuss hard finish plasters and cement plaster.

(b) How steel is manufactured by Bessemer process. 4

Q7. (Marks 4) (a) Discuss structure of ethyne on the basis of sp-hybridization.

(b) Convert ethyl bromide into : (i) Ethane. (ii) Propane. (iii) Ethyl alcohol (iv) n-Butane 4

Q8. (Marks 4) (a) Give the reaction of ethene with (i) H2SO4 (ii) O3 (iii) HOX (iv) Br2

(b) Describe the various tests for identification of carboxyl compounds (Any four): 4

Q9. (Marks 4) (a) Write the mechanism of nitration and sulphonation of benzene.

(b) Describe the two reactions of alcohol and phenol in which "O-H" bond break. 4